Ojas Patil

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Education

University of Wisconsin-Madison Madison, WI
B.S. in Computer Engineering, Computer Science *Sep 2024 – May 2028*

- **Relevant Coursework:** Digital System Design & Synthesis, Machine Organization & Programming, Data Structures & Algorithms
- **GPA:** 3.94/4.0
- **Activities:** Wisconsin Space Program - Avionics Team
- **Honors:** Dean's Honor List (3/3 Semesters)

Technical Skills

Programming Languages: SystemVerilog, Java, MATLAB, C/C++, Assembly Code, HTML/CSS

CAD & Simulation Tools: KiCad, AutoCad Fusion, ModelSim, QuestaSim, LTspice, Altium Designer

Libraries & Tools: Linux, Git, Bash, Latex

Projects

LED Launch Signature System Sep 2025 – Oct 2025
Student Org Project - Wisconsin Space Program *PCB Design, C++ Programming*

- Designed a PCB using Altium Designer to create a launch signature system for a high-altitude rocket.
- Programmed an Arduino Nano microcontroller in C++ to control the LED patterns based on the rocket's flight phases.
- Integrated with the rocket's existing systems to ensure reliable operation during launch and recovery.

Firefly Jul 2025 – Present
Open-Source Low-Cost Microdrone *C++ Programming, PCB Design, Robotics*

- Developed a ESP8266-based micro-drone with auto levelling capabilities using PID control.
- Designed a flight controller PCB in KiCad and relevant firmware in Arduino C++.
- Repository: github.com/Firefly-MicroDrone

OP Keyboard Mar 2025 – Apr 2025
DIY Mechanical Keyboard *PCB Design, CAD*

- Designed a mechanical keyboard based on the Raspberry Pi Pico that features hot-swappable switches and a knob.
- Designed the PCB in KiCad and the case in Fusion 360.
- Uses the QMK firmware with support for customizable keymaps and layers using VIA (in-browser app).
- Repository: github.com/SirAxolott/opkeyboard

FPGA-Based Machine Learning Motion Detection System Jan 2025 – May 2025
Course Project - Digital System Fundamentals *System Verilog, FPGA Implementation*

- Implemented a machine learning program to classify a user's motion by processing data from an accelerometer.
- Designed using System Verilog, simulated on ModelSim, compiled on Questa, and tested on an Intel Altera FPGA.
- Based on a right-shift register that captures the 4-bit serial data packet.
- Repository: github.com/SirAxolott/ECE352-AHW

Work Experience

Student Stocker Nov 2024 – Present
University of Wisconsin-Madison *Madison, WI*

- Collaborated with team members to efficiently assemble and fulfill ready-to-eat meal package orders for university dining services.
- Managed raw material inventory by verifying incoming shipments and organizing stock in designated storage areas.
- Worked with the catering team to set up manage, and maintain professional food buffets for university events.